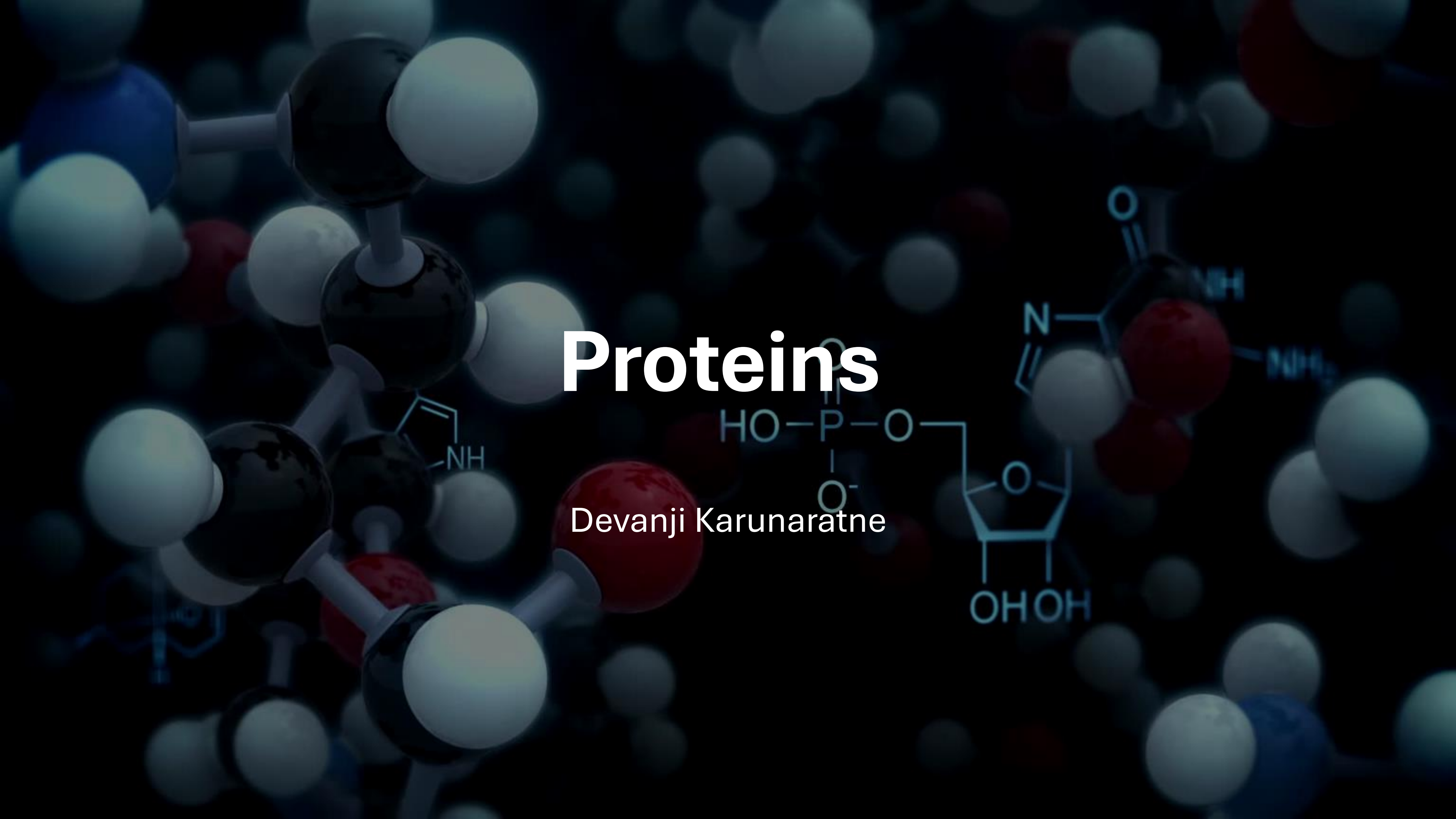
# Proteins

Devanji Karunaratne

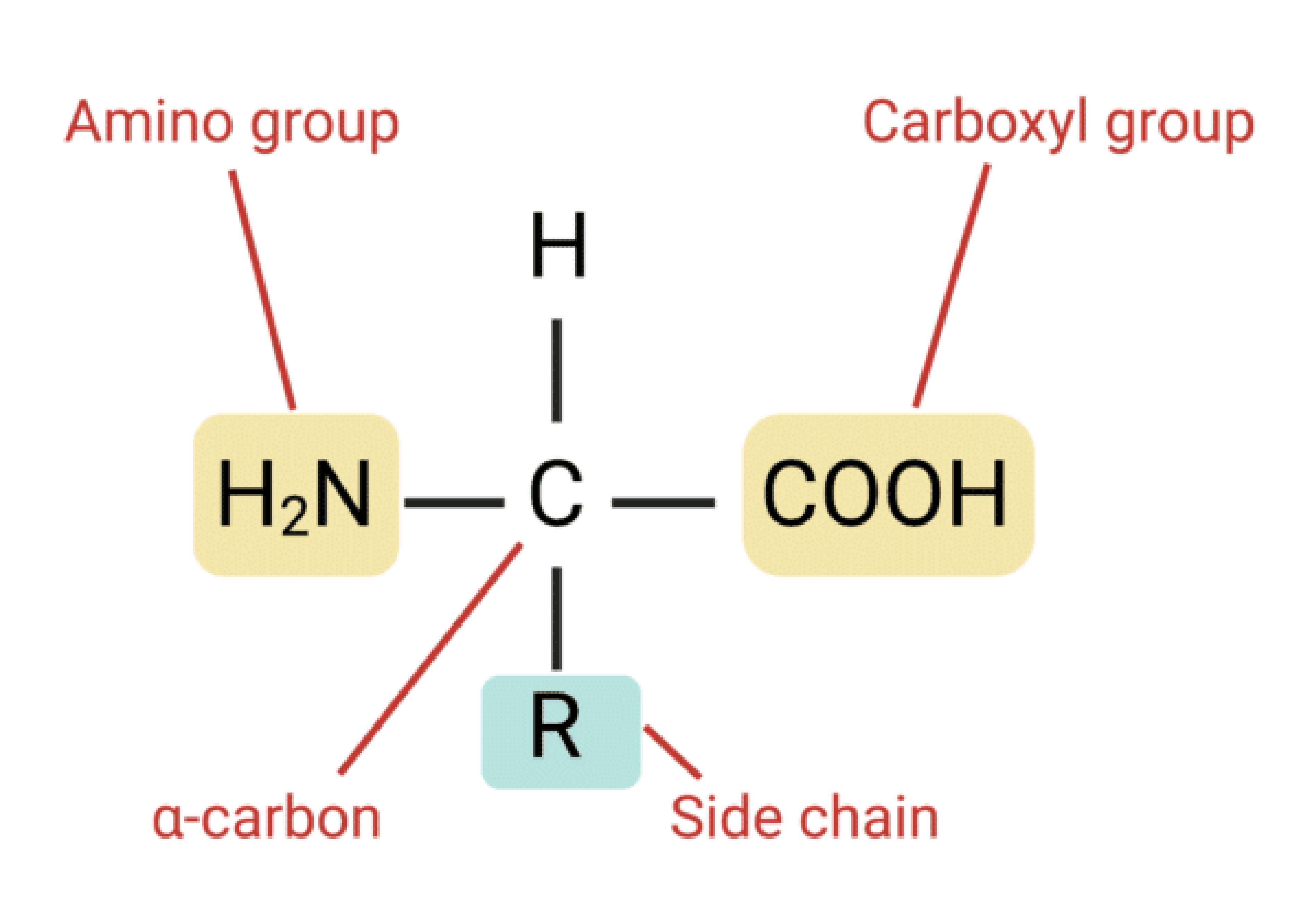


# The Monomers of Proteins

- The fundamental building blocks of proteins are **amino acids**.
- Amino acids are the monomers that link together to form long chains called polypeptides, which then fold into functional proteins.
- They catalyze reactions (enzymes) to transport molecules, provide structural support, and enable cell signaling.

# General Structure of Amino Acids

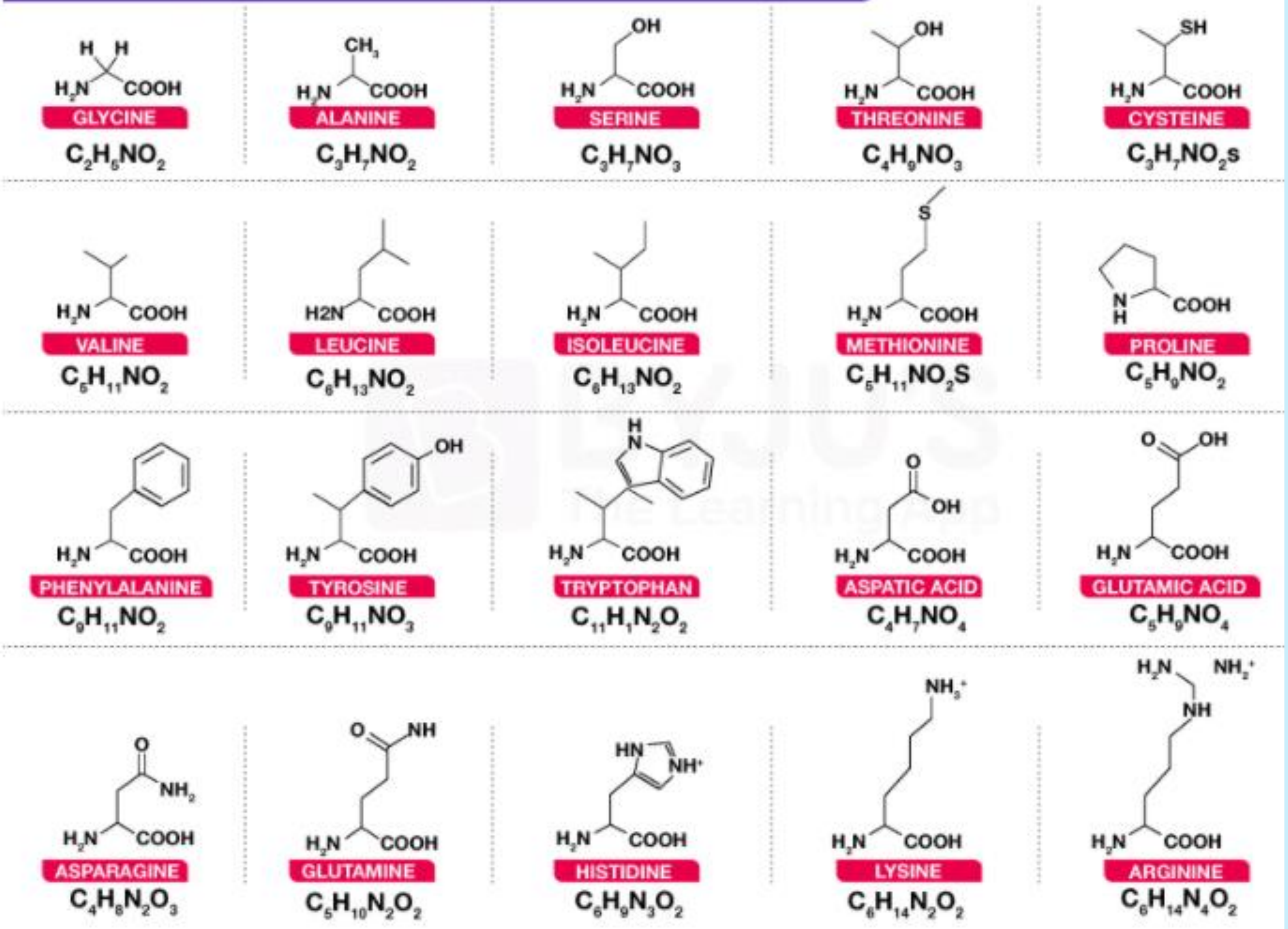
- **Central Carbon Atom (α-carbon):** This carbon is central to the amino acid structure.
- **Amino Group (−$NH_2$):** An amine group attached to the α-carbon.
- **Carboxyl Group (−COOH):** A carboxylic acid group attached to the α-carbon.
- **Hydrogen Atom (−H):** A single hydrogen atom attached to the α-carbon.
- **Side Chain (R-group):** This is the unique part of each amino acid. It varies in structure, size, and electrical charge, and it determines the specific properties and identity of each amino acid.


Amino group
Carboxyl group
H
$H_2N$
C
COOH
R
α-carbon
Side chain

# The α-Carbon as a Chiral Center

- With four different groups attached (except for glycine, where the R-group is also an H atom), the α-carbon is a **chiral center** for 19 of the 20 common amino acids.
- This means amino acids exist as stereoisomers (enantiomers) that are non-superimposable mirror images of each other.
- In biological systems, amino acids almost exclusively exist in the **L-configuration**. (D-amino acids are rare but found in bacterial cell walls and some antibiotics).

# STRUCTURE OF AMINO ACIDS



# Properties and Classification of Amino Acids

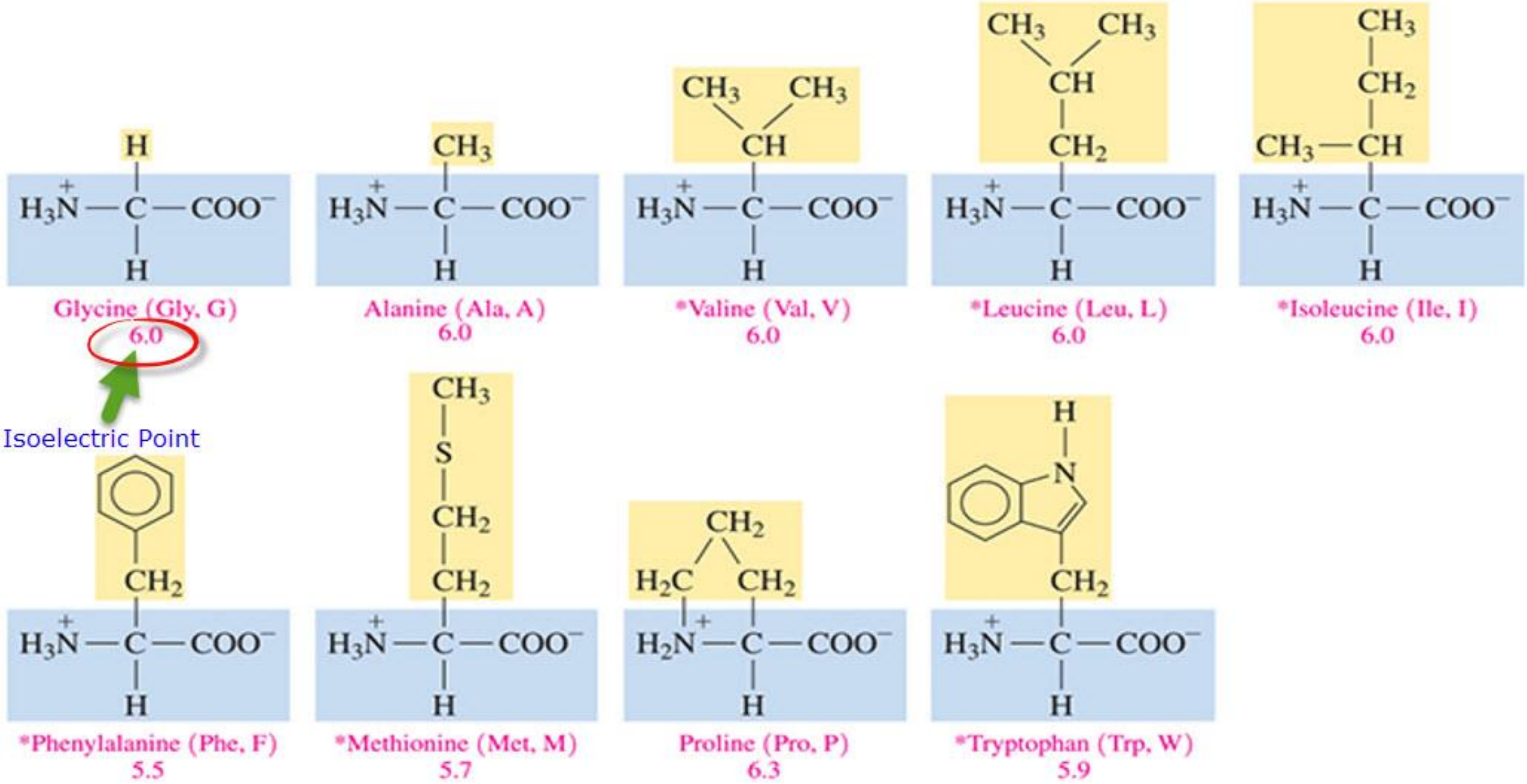
- The unique properties of proteins largely stem from the diverse chemical properties of their constituent amino acid side chains (R-groups).
- We classify amino acids based on the characteristics of their R-groups, particularly their polarity and charge at physiological pH (around 7.4).

# Nonpolar (Hydrophobic) Amino Acids

- These amino acids have R-groups that are composed primarily of hydrocarbons (alkyl chains or aromatic rings).
- They are hydrophobic, meaning they tend to avoid water and are often found in the interior of soluble proteins, away from the aqueous environment, or within the lipid bilayer of membrane proteins.

**Examples:**

- Glycine (Gly, G), Alanine (Ala, A)
- Valine (Val, V), Leucine (Leu, L)
- Isoleucine (Ile, I), Methionine (Met, M)
- Proline (Pro, P), Phenylalanine (Phe, F)
- Tryptophan (Trp, W)



Glycine (Gly, G)
6.0
Alanine (Ala, A)
6.0
*Valine (Val, V)
6.0
*Leucine (Leu, L)
6.0
*Isoleucine (Ile, I)
6.0
Isoelectric Point
*Phenylalanine (Phe, F)
5.5
*Methionine (Met, M)
5.7
Proline (Pro, P)
6.3
*Tryptophan (Trp, W)
5.9

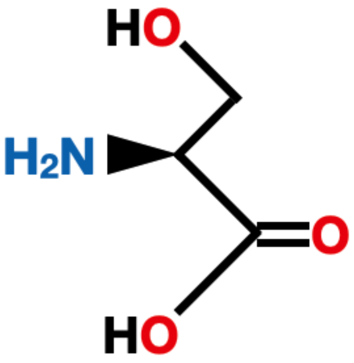
# Uncharged Polar Amino Acids

- These amino acids have R-groups that contain functional groups capable of hydrogen bonding (e.g., hydroxyl, sulfhydryl, amide groups).
- They are hydrophilic and tend to be found on the surface of soluble proteins.

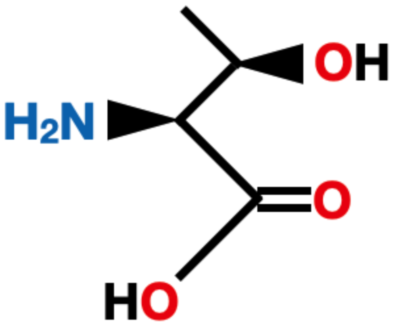
**Examples:**

- Serine (Ser, S)
- Threonine (Thr, T)
- Cysteine (Cys, C)
- Tyrosine (Tyr, Y)
- Asparagine (Asn, N)
- Glutamine (Gln, Q)

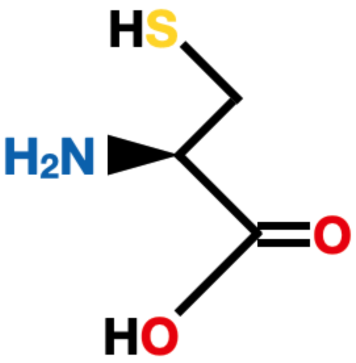
# Amino Acids with Polar Uncharged Side Chains



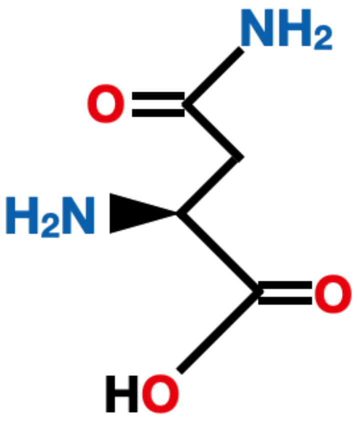
Serine
Ser
S



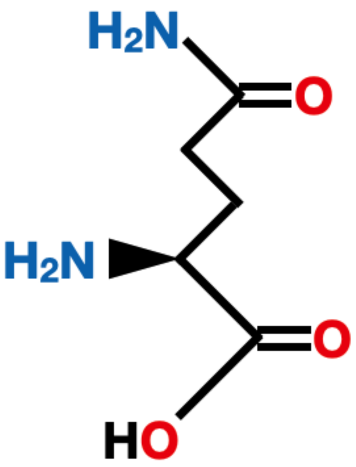
Threonine
Thr
T



Cysteine
Cys
C

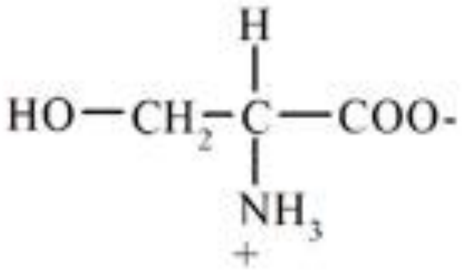


Asparagine
Asn
N

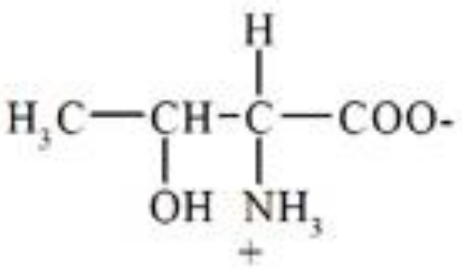


Glutamine
Gln
Q

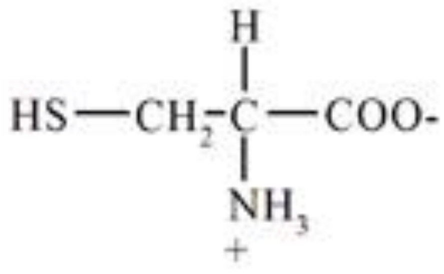
**Group 2 – Amino acids with uncharged polar R groups**



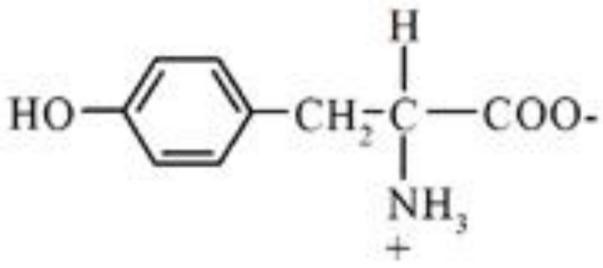
Serine
(Ser)



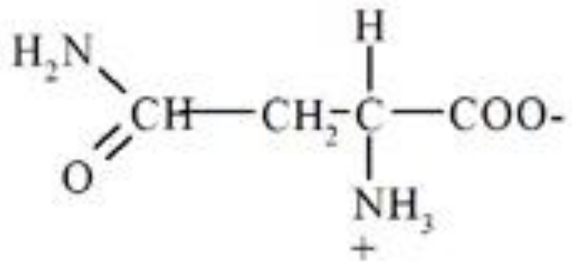
Threonine
(Thr)



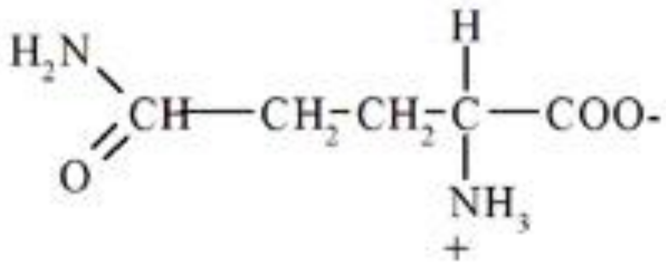
Cysteine
(Cys)



Tyrosine
(Tyr)



Asparagine
(Asn)



Glutamine
(Gln)

# Charged Polar Amino Acids

These amino acids have R-groups that contain ionizable groups, meaning they can carry a net charge (positive or negative) at physiological pH. They are highly hydrophilic and are almost always found on the surface of proteins, involved in ionic interactions, or in enzyme active sites.

## Positively Charged (Basic) Amino Acids:

Have an extra amino group in their side chain that can accept a proton.

Examples: Lysine (Lys, K), Arginine (Arg, R), Histidine (His, H)

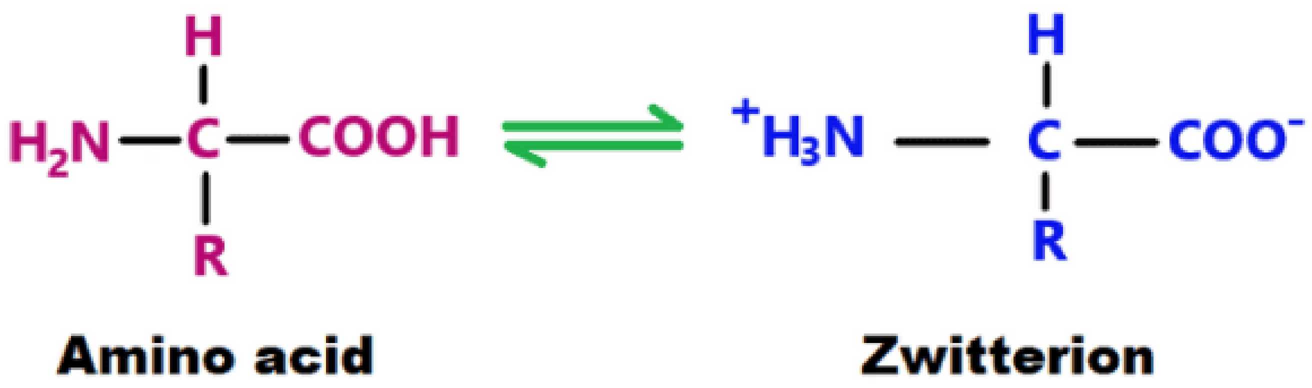
## Negatively Charged (Acidic) Amino Acids:

Have an extra carboxyl group in their side chain that can donate a proton.

Examples: Aspartate (Asp, D), Glutamate (Glu, E)

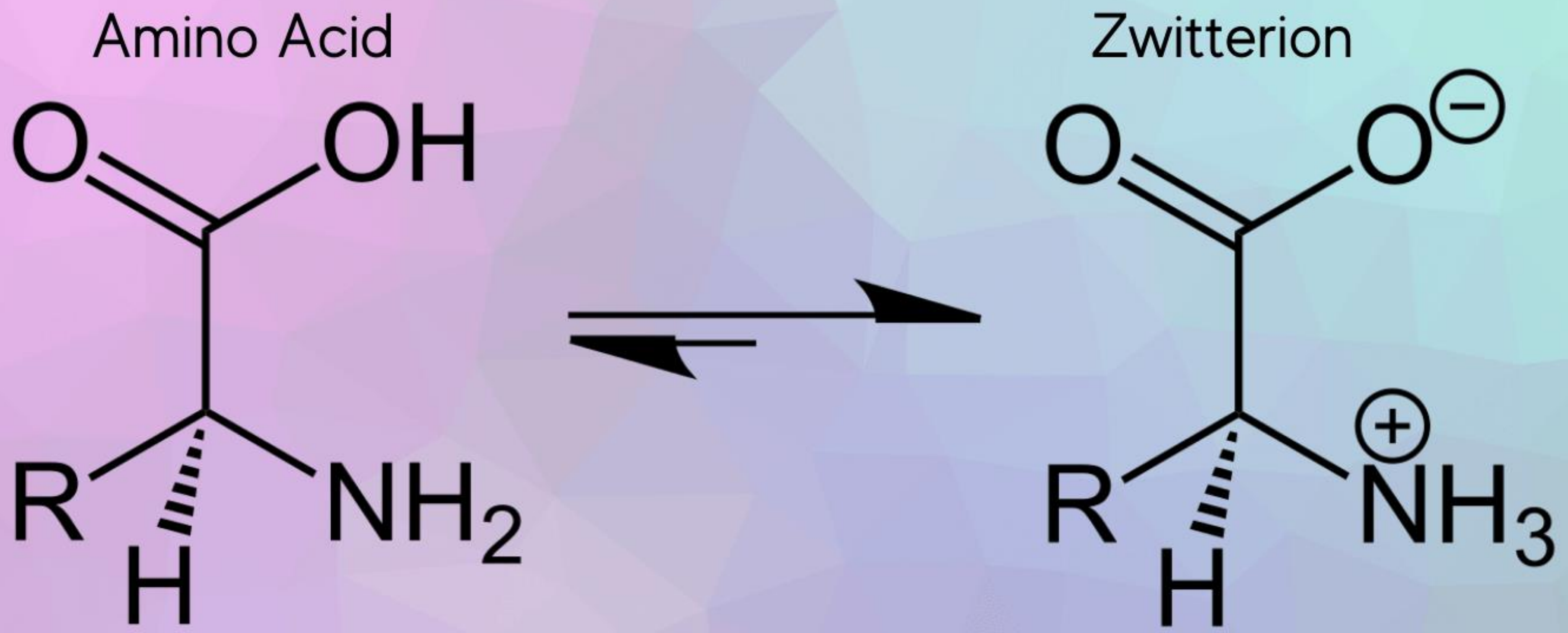
# Ionization Properties of Amino Acids (Zwitterions)

- Amino acids are **amphoteric** molecules, meaning they can act as both acids (donating protons) and bases (accepting protons).
- At physiological pH (around 7.4), the amino group is typically protonated (−NH3+) and the carboxyl group is deprotonated (−COO−).
- This creates a molecule with both a positive and a negative charge, but a **net charge of zero**.
- Such a molecule is called a **zwitterion** (German for "hybrid ion").



# Zwitterion

A zwitterion is a molecule with a net neutral charge that has positive and negative charged functional groups.

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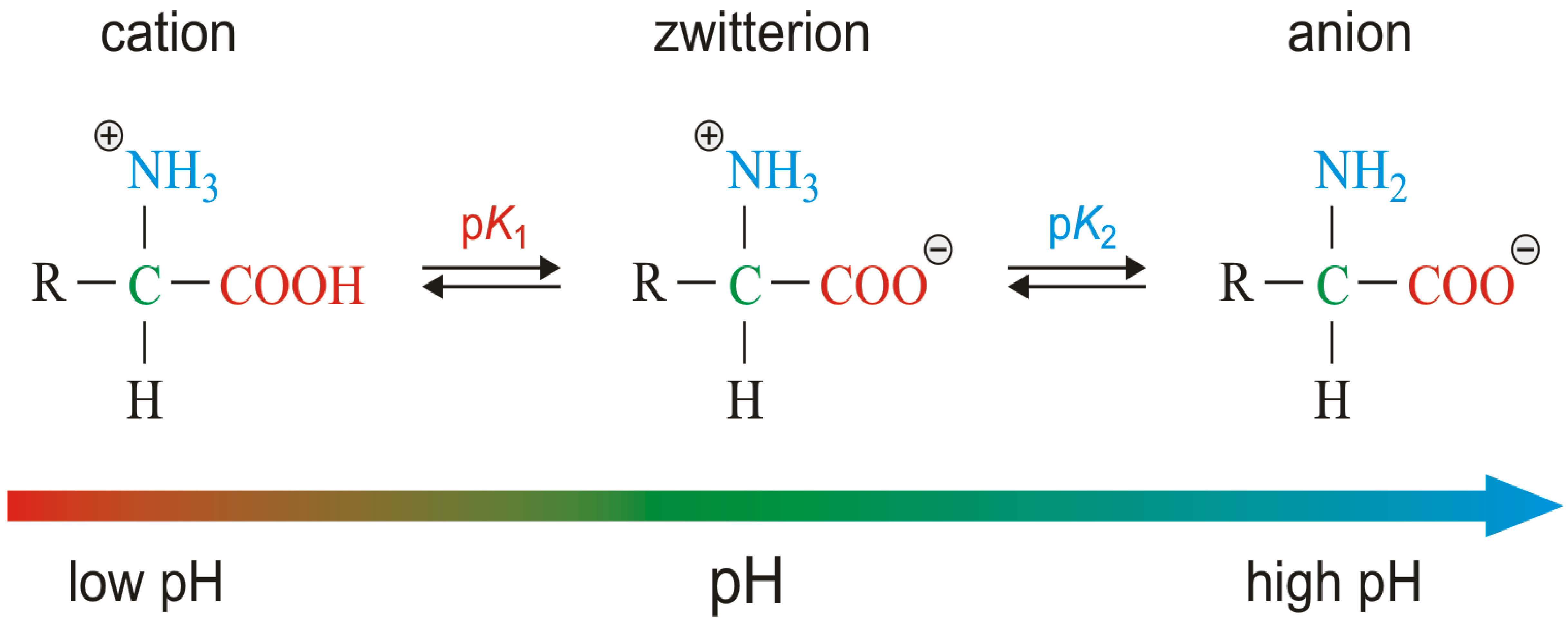
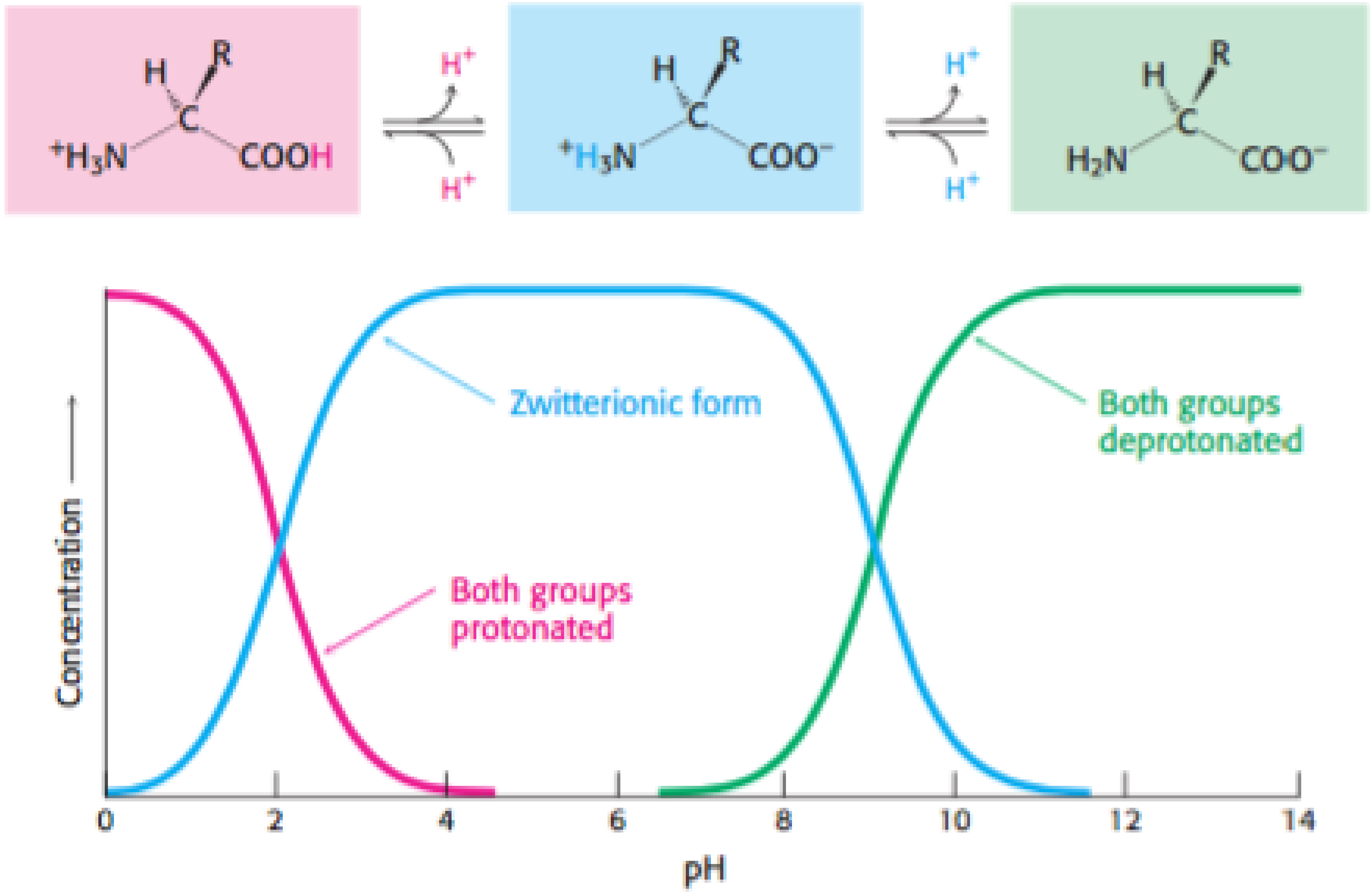
- **Titration Curves:** The ionization states of amino acids change with pH, which can be visualized through a titration curve.
  - At very low pH (acidic), both the amino and carboxyl groups (and any ionizable R-group) will be protonated.
  - As pH increases, the most acidic group (carboxyl) will deprotonate first, followed by the amino group, and then any ionizable R-group.
  - **pKa:** The pH at which 50% of an ionizable group is in its protonated form and 50% is in its deprotonated form.
  - **Isoelectric Point (pI):** The specific pH at which an amino acid (or protein) has a **net electrical charge of zero**. For amino acids with non-ionizable R-groups, the pI is typically the average of the pKa values of the α-carboxyl and α-amino groups. For amino acids with ionizable R-groups, the pI is calculated differently, taking into account all three pKa values.


cation
zwitterion
anion
pK1
pK2
low pH
pH
high pH



H
R
C
$^+H_3N$
COOH
$H^+$
$H^+$
H
R
C
$^+H_3N$
$COO^-$
$H^+$
$H^+$
H
R
C
$H_2N$
$COO^-$
Zwitterionic form
Both groups deprotonated
Both groups protonated
Concentration
0
2
4
6
8
10
12
14
pH

# Reactions of Amino Acids

## Peptide Bond Formation:

- It's a **condensation reaction** (dehydration synthesis) between the **carboxyl group** of one amino acid and the **amino group** of another amino acid, with the elimination of a water molecule.
- The resulting covalent bond is called a **peptide bond** (or amide bond).

## Disulfide Bond Formation:

- Specific to **cysteine** residues.
- The sulfhydryl (-SH) group of two cysteine residues can undergo oxidation to form a **disulfide bond** (−S−S−).

# Reactions of Amino Acids

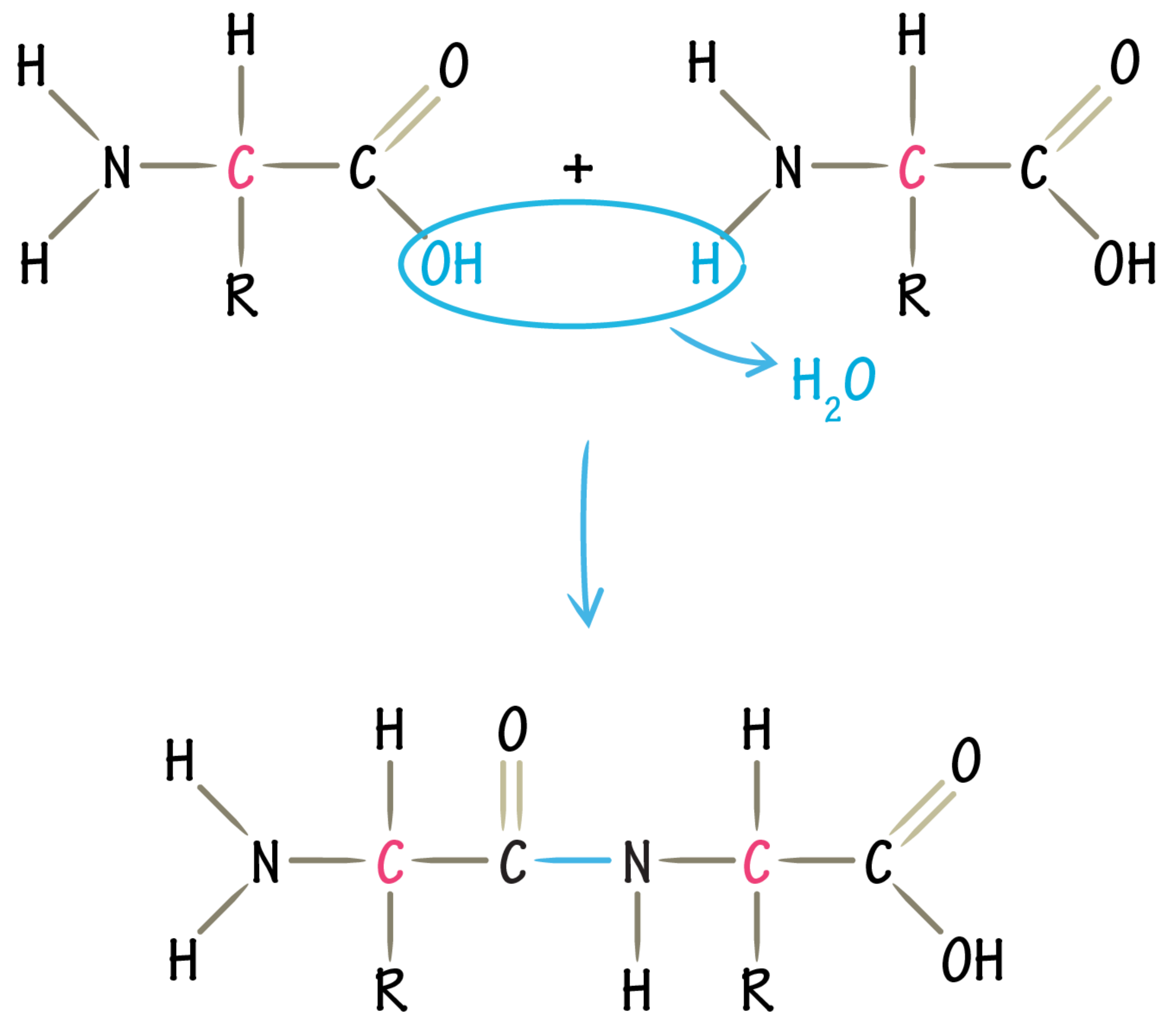
## Reactions of Amino and Carboxyl Groups (beyond peptide bond):

- Both amino and carboxyl groups can participate in other reactions, such as **esterification** (carboxyl + alcohol) or **acylation** (amino + acyl group).

## Post-Translational Modifications of R-groups:

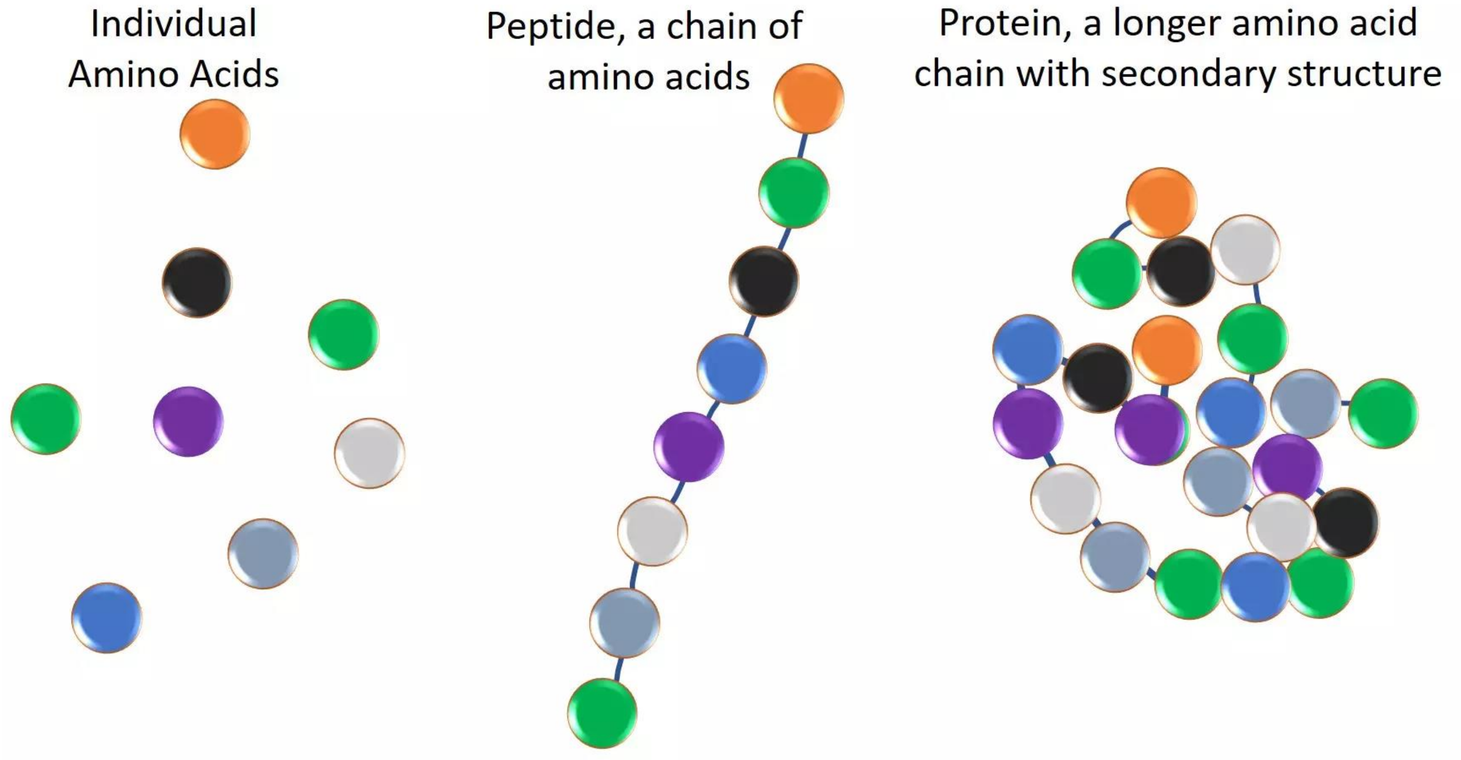
- **Phosphorylation:** Addition of a phosphate group
- **Glycosylation:** Attachment of carbohydrate chains
- **Acetylation:** Addition of an acetyl group
- **Hydroxylation:** Addition of a hydroxyl group
- **Methylation:** Addition of methyl groups

# Peptide Bond Formation



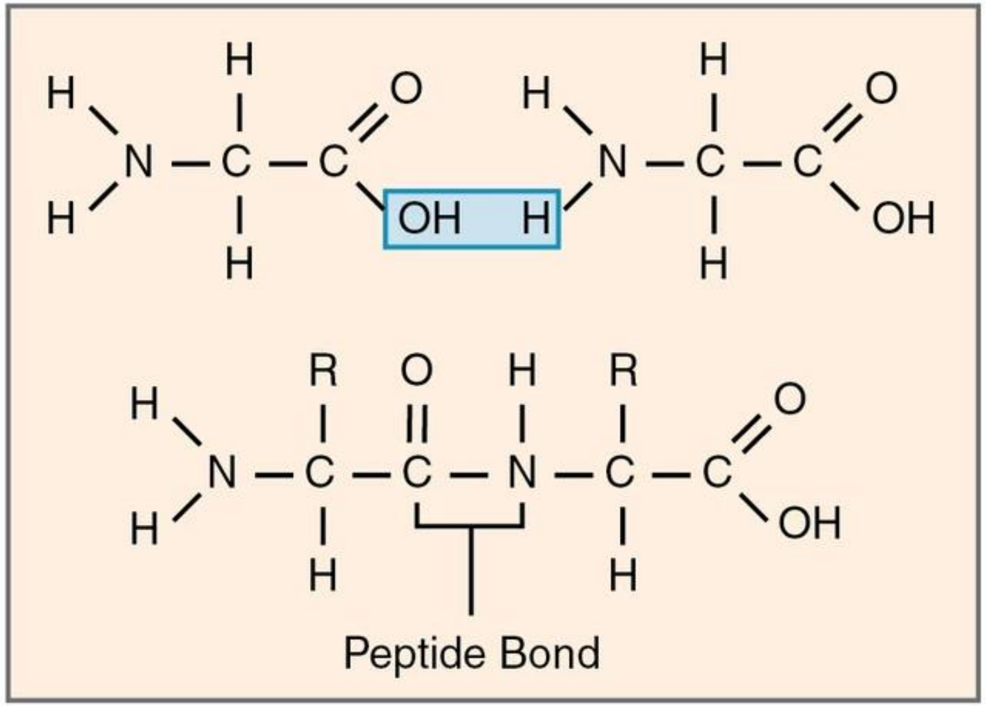
# Peptides: Chains of Amino Acids

- A **peptide** is a short chain of amino acids linked by peptide bonds.
- A **polypeptide** is a longer chain (often arbitrarily defined as >50 amino acids), and a **protein** is a functional polypeptide (or multiple polypeptides) that has folded into a specific 3D structure.



# Structure of Peptides

**Peptide Bond:** The defining feature. It has partial double-bond character due to resonance, making it rigid and planar. This limits rotation around the peptide bond itself.



# Structure of Peptides

**N-terminus and C-terminus:**

- Every polypeptide chain has a free amino group at one end (the **N-terminus** or amino terminus) and a free carboxyl group at the other end (the **C-terminus** or carboxyl terminus).
- Proteins are synthesized from the N-terminus to the C-terminus.

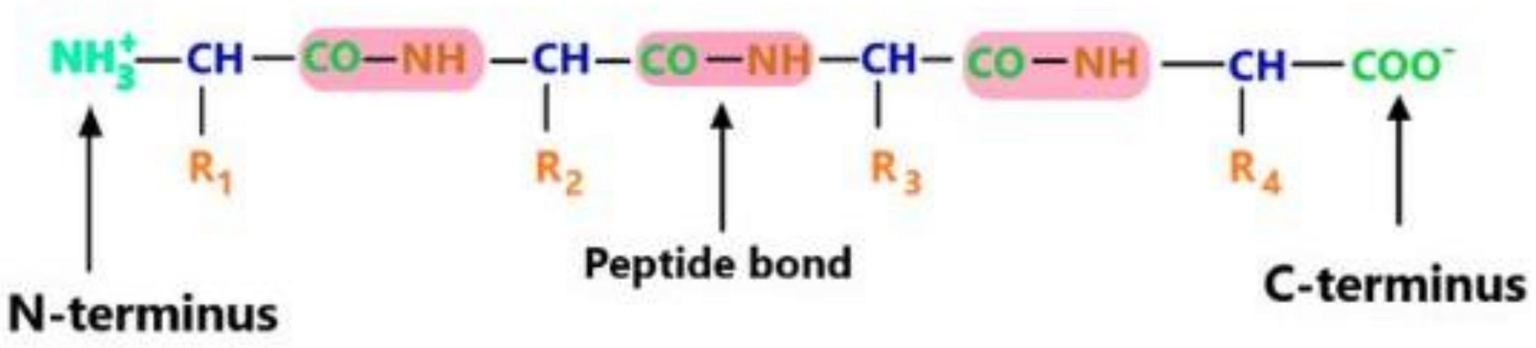
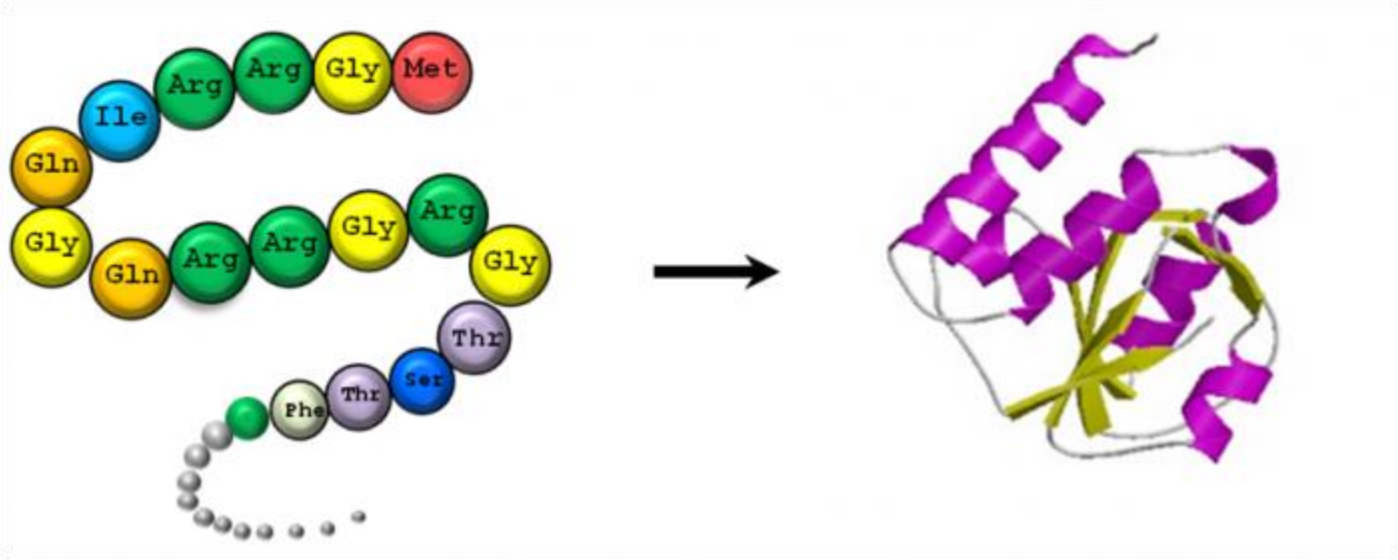



FIG: N-terminus and C-terminus of polypeptide chain

# Structure of Peptides

## Amino Acid Sequence (Primary Structure):

- The unique, linear order of amino acids in a polypeptide chain is its **primary structure**.
- It dictates how the polypeptide will fold into its specific 3D structure and thus determines its function.



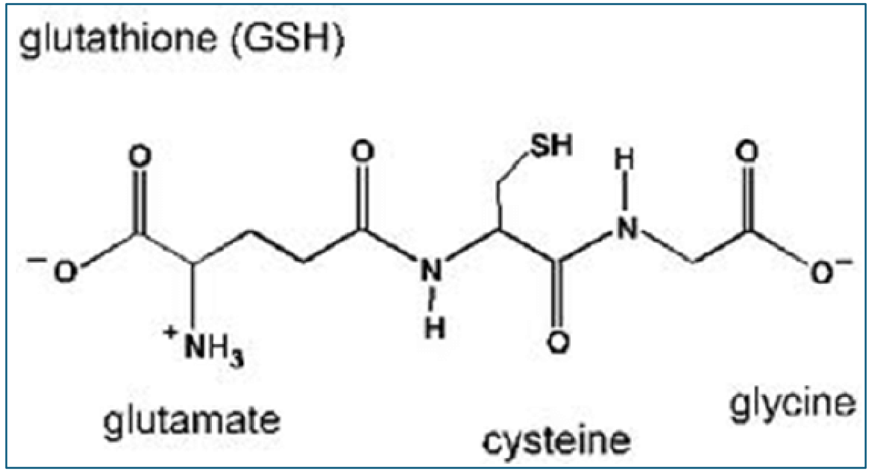
# Examples of Biologically Active Peptides

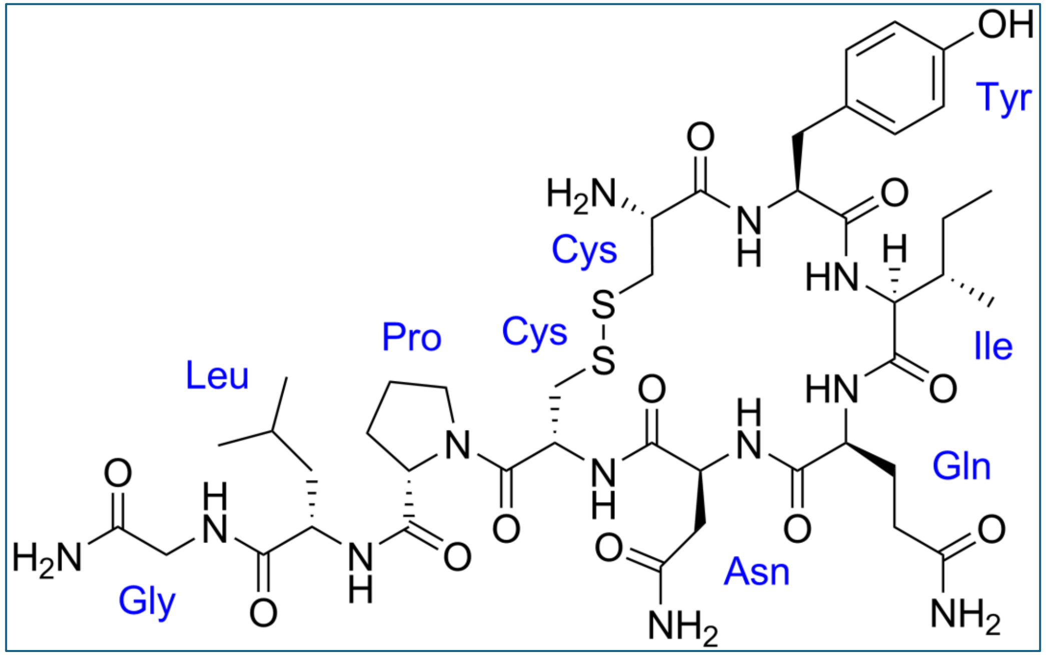
## Dipeptides:

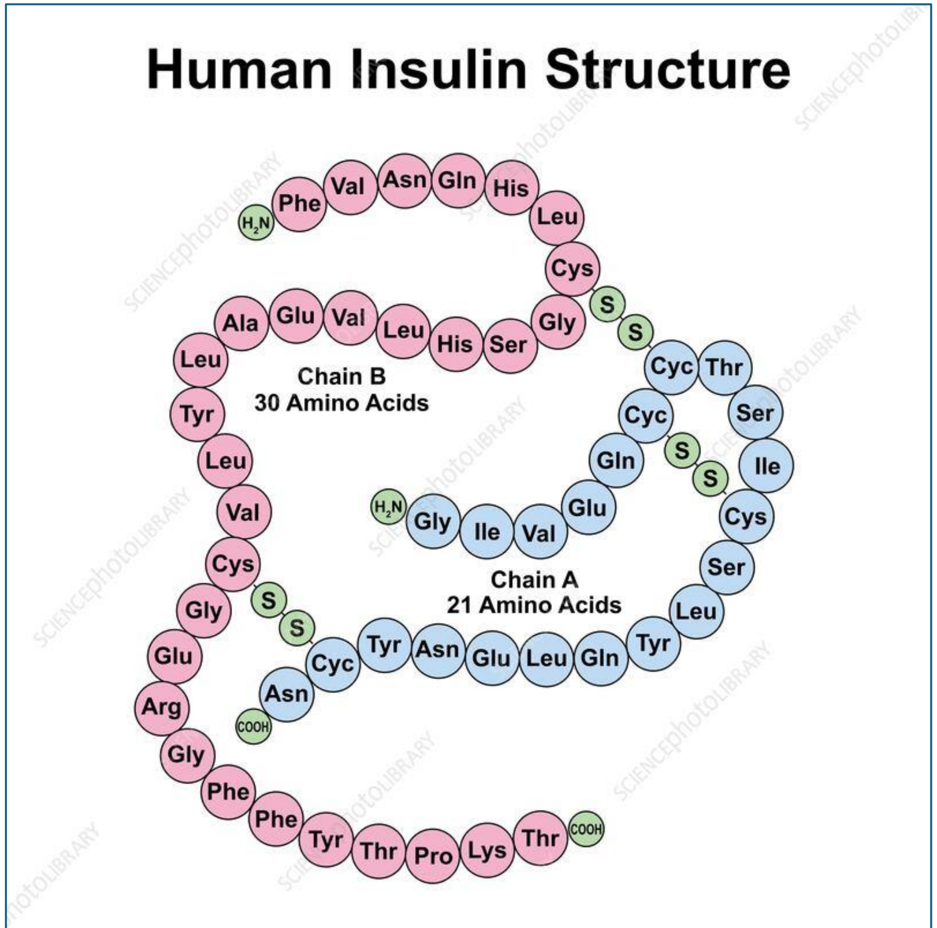
- **Carnosine:** Found in muscle and brain tissue, acts as an antioxidant and pH buffer.
- **Aspartame:** An artificial sweetener.

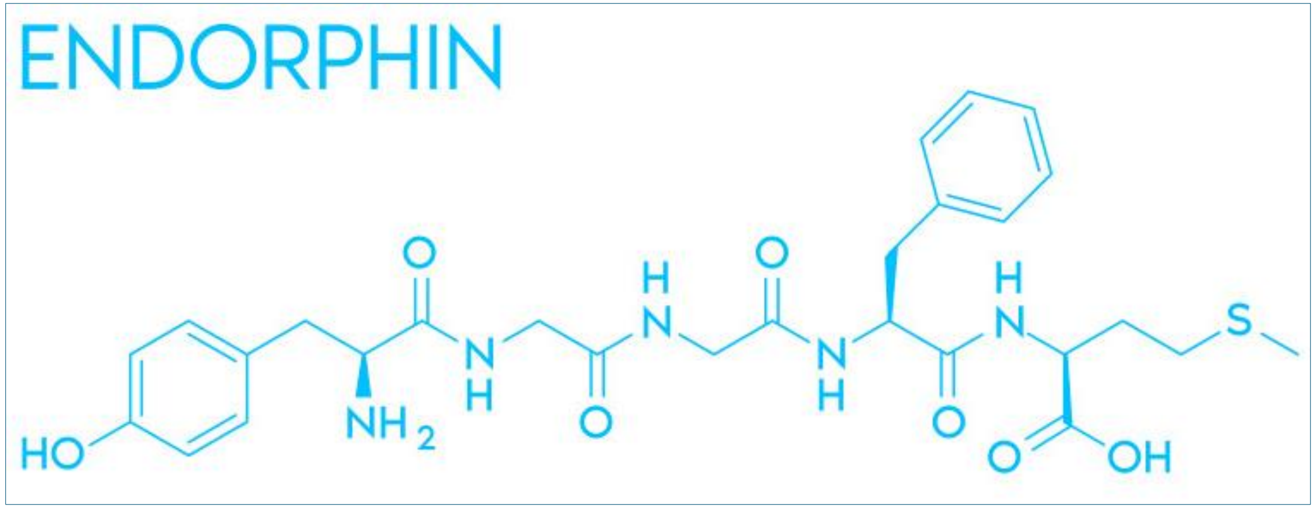
## Oligopeptides:

- **Glutathione:** A tripeptide. Major antioxidant in cells, involved in detoxification.
- **Oxytocin:** (9 amino acids) A hormone involved in uterine contractions and milk ejection.
- **Vasopressin (Antidiuretic Hormone, ADH):** (9 amino acids) Regulates water reabsorption in the kidneys.
- **Insulin:** (51 amino acids, two chains linked by disulfide bonds) A hormone that regulates blood glucose levels.
- **Bradykinin:** (9 amino acids) A potent vasodilator involved in inflammation.
- **Endorphins/Enkephalins:** Short peptides acting as natural pain relievers and neurotransmitters.


glutathione (GSH)
glutamate
cysteine
glycine


Tyr
Cys
Ile
Pro
Cys
Leu
Gln
Gly
Asn


Human Insulin Structure
Chain B
30 Amino Acids
Chain A
21 Amino Acids

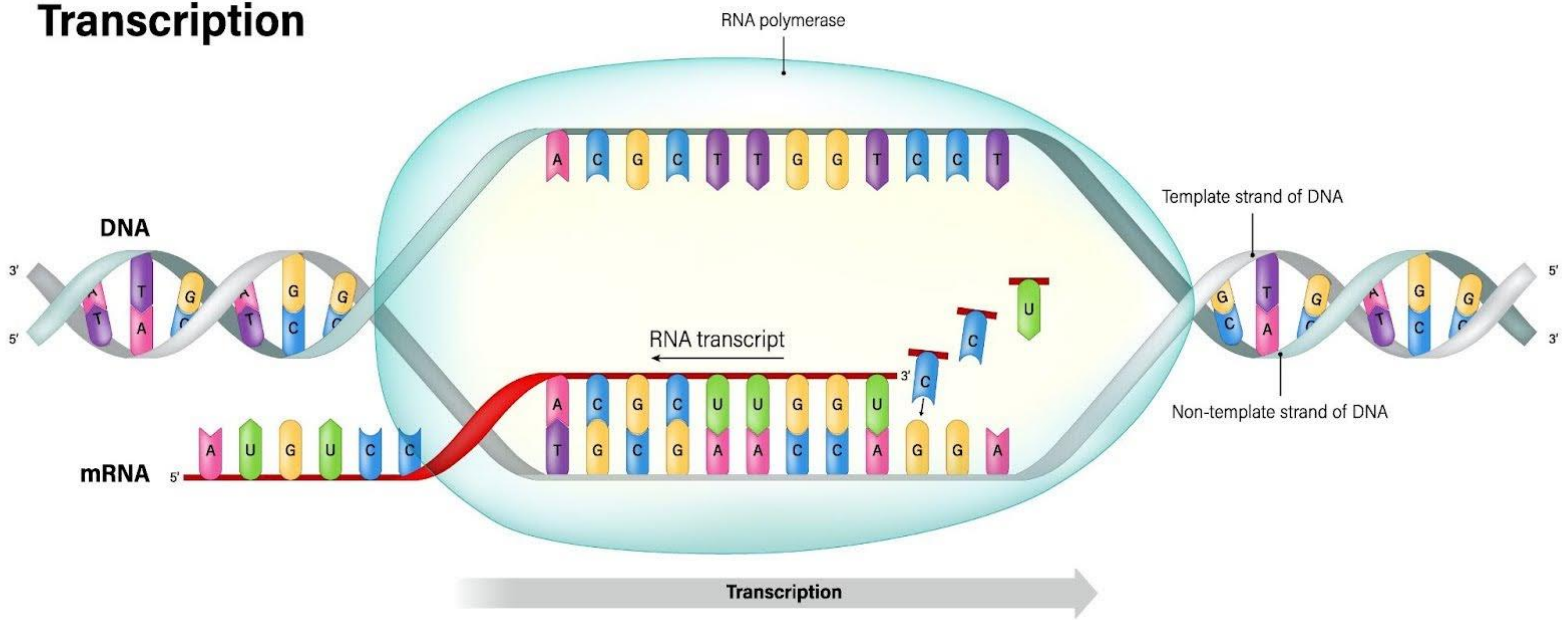

ENDORPHIN

# Synthesis of Peptides

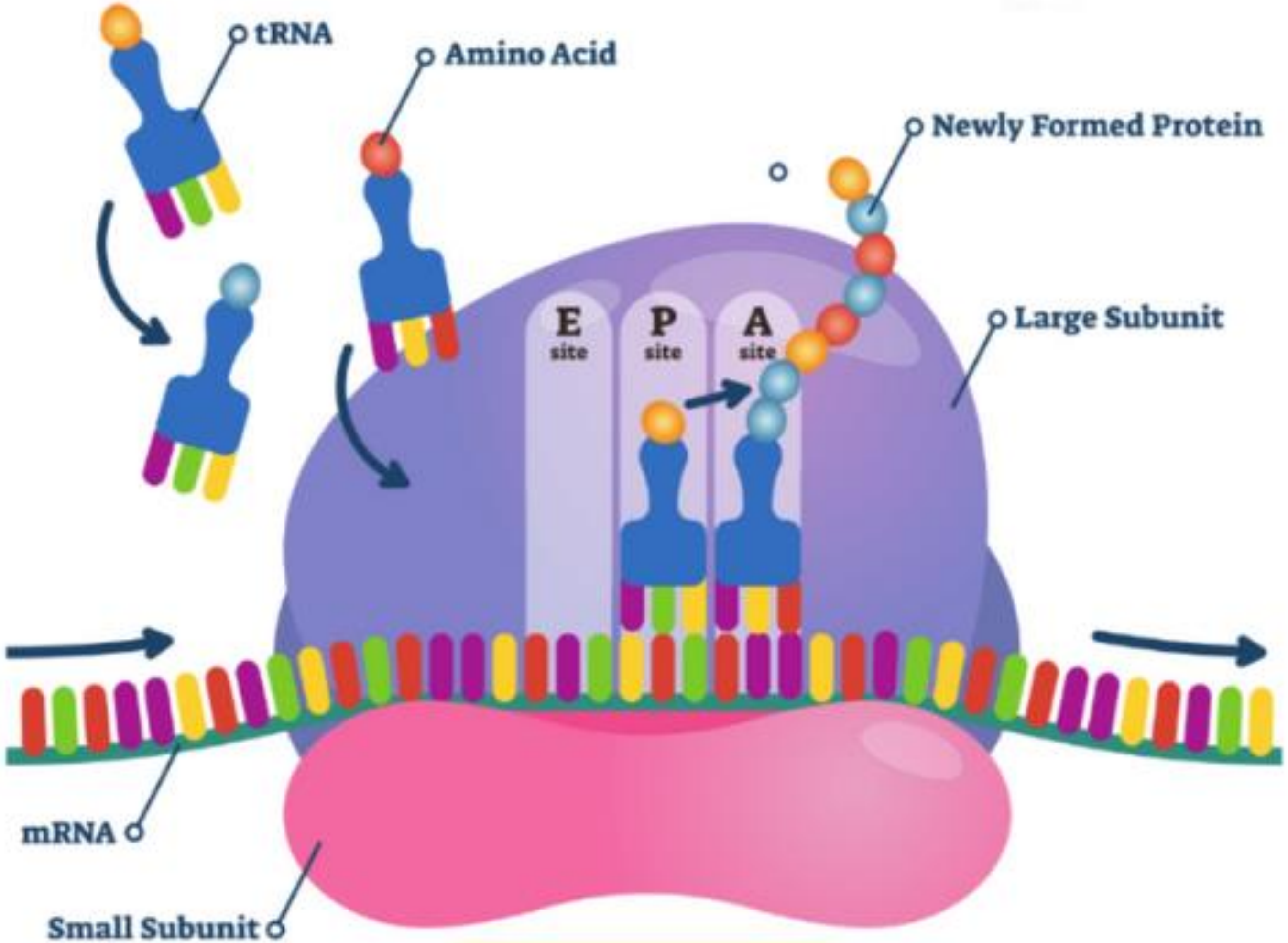
## In Vivo Synthesis (Ribosomal Protein Synthesis / Translation):

- **Template:** Messenger RNA (mRNA) carries the genetic code from DNA.
- **Machinery:** Ribosomes (complexes of ribosomal RNA and proteins) are the sites of protein synthesis.
- **Reactants:** Transfer RNA (tRNA) molecules bring specific amino acids to the ribosome, guided by the codons on the mRNA.
- **Mechanism:** Amino acids are added one by one to the growing polypeptide chain, starting from the N-terminus. A peptide bond is formed between the carboxyl group of the last amino acid in the chain and the amino group of the incoming amino acid. This reaction is catalyzed by the ribosomal machinery itself (peptidyl transferase activity).
- **Directionality:** Always proceeds from the N-terminus to the C-terminus.

# Transcription



# Protein Translation

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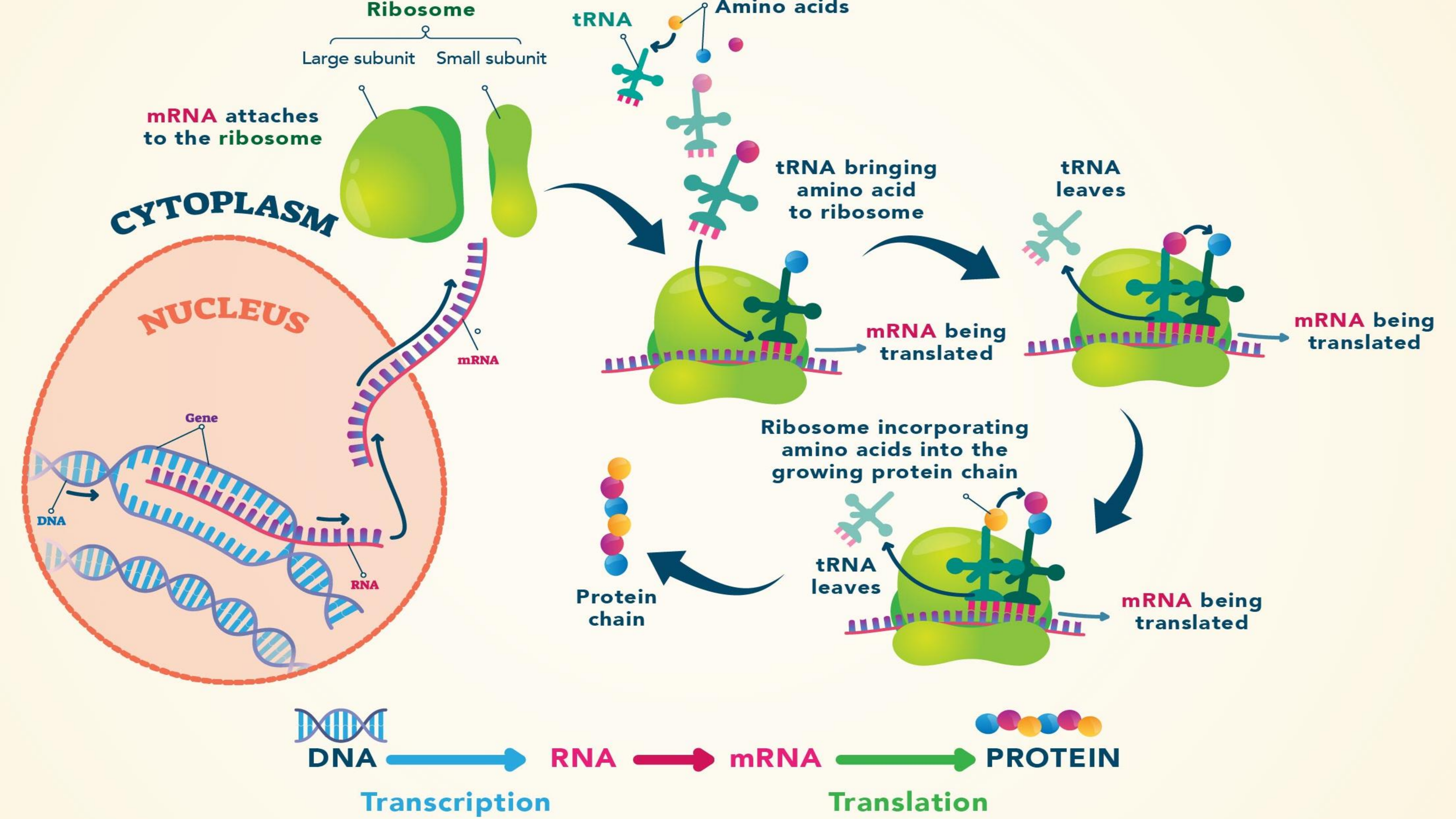

Ribosome
Large subunit
Small subunit
tRNA
Amino acids
mRNA attaches to the ribosome
CYTOPLASM
NUCLEUS
mRNA
Gene
DNA
RNA
tRNA bringing amino acid to ribosome
mRNA being translated
tRNA leaves
mRNA being translated
Ribosome incorporating amino acids into the growing protein chain
tRNA leaves
mRNA being translated
Protein chain
DNA
RNA
mRNA
PROTEIN
Transcription
Translation

THANK YOU